Indian Institute of Technology, Bhilai

CSP203: Software Tools & Technologies Laboratory

Assignment - 2

Full Marks: 50 *Deadline: 11:59 PM, 27-08-2025*

Submission Instructions:

1) Answer all the questions.
2) Write all the scripts, execute and take screenshots of the scripts, name the scripts files as <Your_Roll_No>_<Lab_Assignment_No>_Assignment_<Question_No>.
3) Prepare a report pdf which contains the question number and script code and corresponding outputs. Report quality will contain **5 marks**. You should prepare a zip folder which will contain the report and all the script files.
4) Naming of the zip file should strictly follow the given format:
   <Roll_No>_<Name>_Assignment_<Assignment_No>
   If your name is Shareen, roll number is B24CS055 then the filename should be:
   B24CS055_Shareen_Assignment_02.pdf
5) **Plagiarism is strictly prohibited.**

Objectives:

1) Getting familiar with bash scripting commands.
2) Using piping, shell commands in script and implementing file and database handling.
3) Learning proper usage of loops, array and conditional statements.
4) Implementing a menu program to experience the importance of scripting.

Task:

Implement a shellATM machine using the shell script with the following features.

1. Account.txt contains several lines in the below format:

   Account Name, Card Number, Email ID, Balance

   Account name: It should contain First name followed by Last name (with space separated). First and Last names start with capital letters followed by any number of lower letter alphabets.

   Card Number: It has 16 digits in the below format:

   <4 Digits><space><4Digits><space><4 Digits><space><4 Digits>

   Email ID: It has the format username@Domain.

   The username should start with an alphabet (lower/upper) followed by any number of alphabet or digits. Domain should use <string>.<string>.<string>, where strings contain only lower Alphabets.

   Balance: Any number >= 0.
2. Credentials.txt has the below format.

   Card Number, Password

   Card number format follows as listed in Account.txt. Passwords can be any string.

Step-0: Your shellATM script upon execution should show a welcome screen as follows.

**** Welcome to My shellATM ****

**** Press any key to continue ***

Step-1: Once any key is pressed, the terminal screen should seek the credentials of Card Number and Password. If both the Card Number and Password match as listed in Credentials.txt, it should authenticate the users. Otherwise, it should ask to enter the details again. [Hint: You can use grep commands here]

Step-2: A successful authenticated user should get the following options.

1. Withdraw cash
2. Deposit cash
3. Settings
4. Exit

Withdraw Cash/Deposit cash options should accept a non-negative number and subtract/add the cash amount entered by the user. Once the operation is complete, your terminal should show the updated balance on the screen. [*Hint: Use grep with regular expression to validate the number and sed for updation of balance*].

Settings: Option should provide the user the user to change the email address. The new account and email address should use the same syntax as listed in the Account.txt. Any invalid entry should throw an error. If the new entries are valid, you should update the Account.txt. [Hint: You can use grep for checking the patterns and sed for replacement]

After completion of any of the options in Step-2, the terminal should go to the welcome screen.

Suggestion:

1. Your shell script should use as many linux commands as possible to minimize the work.
2. You can use read to get the input from the user.